Supplementary Information

A catalog of noncanonical HTH and RHH DNA-binding domain families in Archaea

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Supplementary Methods

Operational scope and terminology

The discovery universe comprises archaeal proteins detected by profiles in Pfam clans CL0123 and CL0057. “Noncanonical” means not explained by a nineteen-family reference set under the stated reciprocal-exclusion rule. It does not mean previously undescribed in all databases, nor does it establish transcription-factor activity.

Panel construction

Assemblies were grouped into seven major lineage categories. All genomes from five underrepresented groups were retained. Euryarchaeota and Crenarchaeota were capped at 140 and 70; subgroup allocations were proportional and genomes were sampled without replacement using seed 42. The final count of 331 was the output of that rule.

Reciprocal exclusion

For each protein, the best-scoring candidate-domain interval was compared with the best canonical hit. Overlap was shared aligned residues divided by candidate-interval length. A candidate was excluded when overlap exceeded 0.5 and the canonical domain score was at least the candidate score minus 5 bits. The highest-scoring retained candidate was used when multiple candidates remained.

Family construction

Retained proteins were clustered with MMseqs2 at minimum identity 0.30 and coverage 0.60, coverage mode 0, cluster mode 1. Clusters required occurrence in at least five genomes, median length below 250 amino acids, at least 60 percent of members below 250 amino acids, and absence of clearly nonregulatory annotation keywords.

Nonregulatory exclusion list

A cluster was excluded when its dominant Pfam label exactly matched or began with one of the following patterns: eIF-5_eIF-2B, Ribosomal_S19e, TFIIE_alpha, SRP19, TFIIE_beta, RNA_pol_Rpb, S10_plectin, Ribosomal_L29, Ribosomal_S13, tRNA_bind, RuvA_N, RuvA_C,

DNA_pol, Rad51, RecA, DEAD, Helicase_C, ResIII, HNH, GIY-YIG, Homeodomain, Pou, Ets, Fork_head, CBFD_NFYB_HMF, Myb_DNA-binding, Myb_DNA-bind, HTH_psq, zf-, Integrase, rve, Recombinase, Resolvase, Terminase, Phage, DNA_binding_1, Sigma54_activat, Sigma70_r2, Sigma70_r3, Sigma70_r1, AAA, Cdc6_C, Mor, CsrA, KH, OB_, RRM, dsrm, PAZ or Piwi. This rule removes obvious non-target families but cannot establish regulatory activity for retained clusters.

Catalog sequence recovery

Protein sequences for all 13,237 released family members were recovered by accession. Versioned NCBI assembly protein FASTA files supplied 13,232 sequences; five accessions absent from their current assembly files were recovered through NCBI Protein. Every recovered sequence matched the length recorded in the membership table. The release includes a deterministic compressed FASTA, source URLs, retrieval dates, source-file checksums, accession-level fallback records and an empty missing-sequence report.

Panel and projection sensitivity

Five hundred seeded within-panel resamples tested three alternative sampling schemes using fixed family membership. A lineage-stratified accumulation analysis sampled 10 to 100% of each major group in 10% increments, with 500 replicates below the complete panel. A separate 500-replicate analysis sampled 331 lineage-matched genomes from the independent 3,192-genome projection matrix after removing discovery-panel accessions. These analyses test fixed-catalog observation, retention and HMM transfer, not de novo rediscovery.

Genome-quality sensitivity

The 3,192 accession-resolved projection genomes were stratified by origin, assembly level, and reported completeness/contamination. Per-family prevalence and detected-family counts were reported. Contrasts are descriptive because strata are confounded with lineage and sampling.

Taxonomic clustered prevalence

Organism names defined 533 inferred genus clusters. Genera were resampled with replacement for 1,000 seeded replicates to obtain genus-balanced and cluster-weighted prevalence intervals. This addresses cluster sampling but is not a species-tree model.

Motif holdout

For twelve families, PWMs were fitted on taxonomically separated training data. Genome-blocked splits held out genomes, genus-blocked splits held out complete genera, and the

strictest analysis left out one major lineage at a time with same-lineage other-family controls. Minimum sample-size failures were reported as unevaluable. Results remain predictive rather than biochemical.

Sequence-family separability

For a prespecified subset of 782 proteins from ten families, normalized dipeptide frequencies were classified by leave-one-sequence-out family centroids. Exact binomial intervals and separation margins were reported. This subset analysis does not validate all 250 families; the complete 13,237-member sequence collection is deposited separately for future catalog-wide validation.

Evidence limits

Neighborhood enrichment is treated as association, motif prediction as sequence-level prediction, and structure matching as similarity to a curated reference set. None alone demonstrates direct DNA binding or transcriptional regulation.

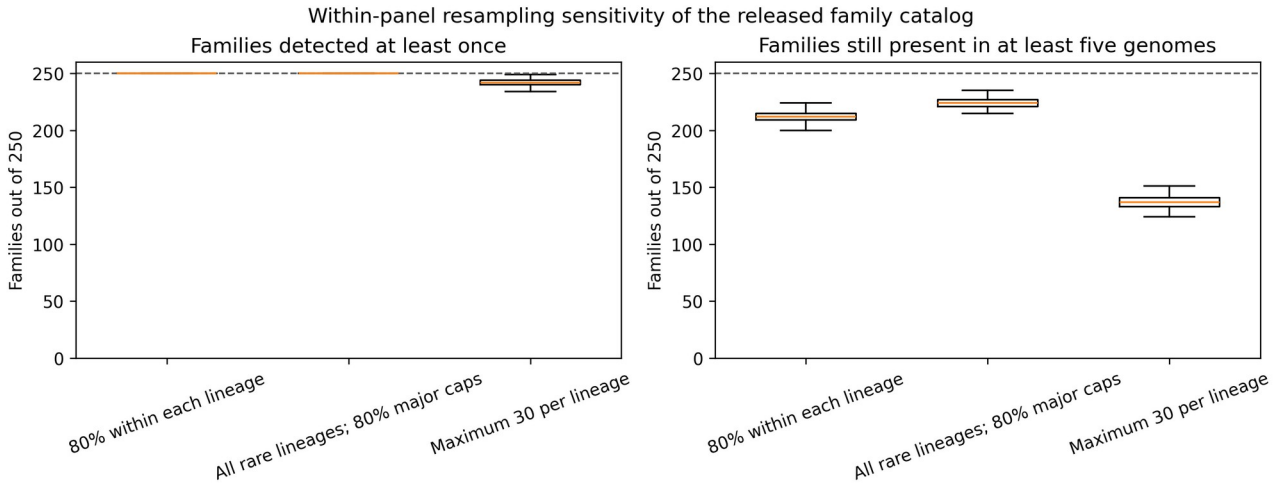
Supplementary Table S1. Analysis classification

Analysis	Role	Question	Principal limit
Catalog size and membership	Primary descriptive	Prespecified released pipeline summaries	Bounded to two Pfam clans
Held-out family HMM assignment	Confirmatory validation	Withheld sequence assignment	Only families with ≥20 members
Independent-genome projection	Confirmatory validation	Zero accession overlap	Fixed HMMs; no rediscovery
Genome/genus/lineage-blocked motif AUC	Confirmatory sensitivity	Out-of-sample discrimination	Predictive, not biochemical
Genome-quality stratification	Sensitivity	Robustness across deposited strata	Descriptive; confounded
Genus-clustered prevalence	Sensitivity	Taxonomic sampling imbalance	Organism-name genera; no species tree
Ten-family sequence separability	Sensitivity	Subset boundary coherence	Only ten families analyzed
Neighborhood enrichment	Exploratory	Functional hypothesis generation	No direct target claim
Haloferax expression	Exploratory	Expression context	Three replicates; simple transformed-count tests
Experimental validation plan	Future work	Direct binding and in vivo regulation	Not performed

Supplementary Table S2. Data artifact definitions

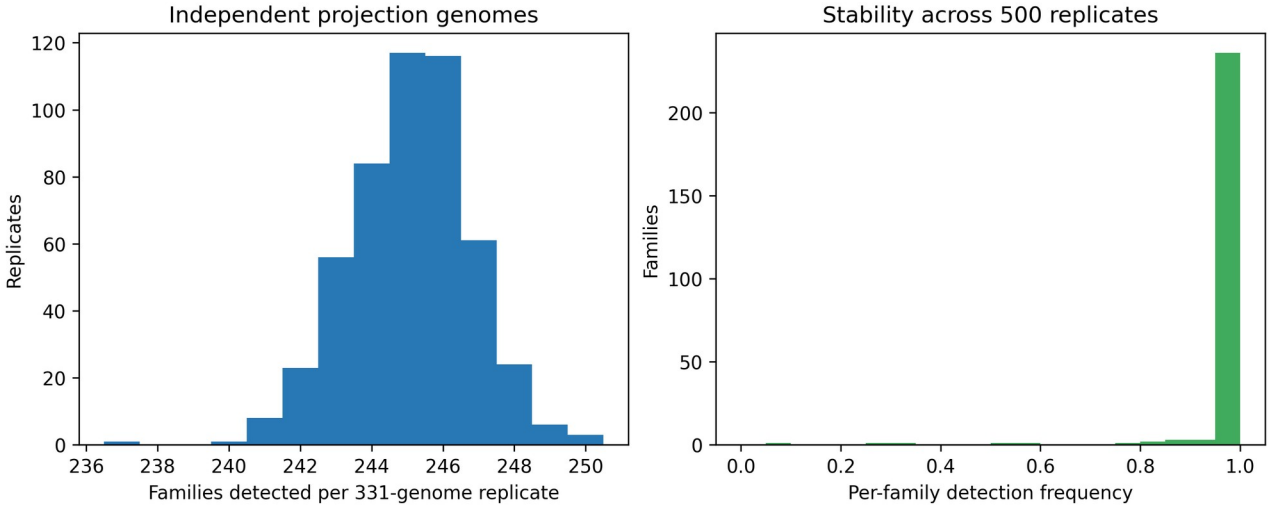
Artifact	Scope	Permitted interpretation
Discovery panel	331 accessions	Family discovery and fixed-member sensitivity
Legacy projection matrix	250 × 3,192	Accession-resolved projection sensitivity
Final genome accounting	3,526 genomes	Final fates and denominators
Corrected prevalence	250 families × 7 groups	Final group-level prevalence only
Family upstreams	12 families; 250-nt sequences	Genome/genus/lineage-blocked motif validation
Phylogenetic sequence subset	782 proteins; 10 families	Subset sequence separability only
Catalog member sequences	13,237 proteins; all 250 families	Sequence-level reanalysis with accessioned recovery provenance

Supplementary Figure S1



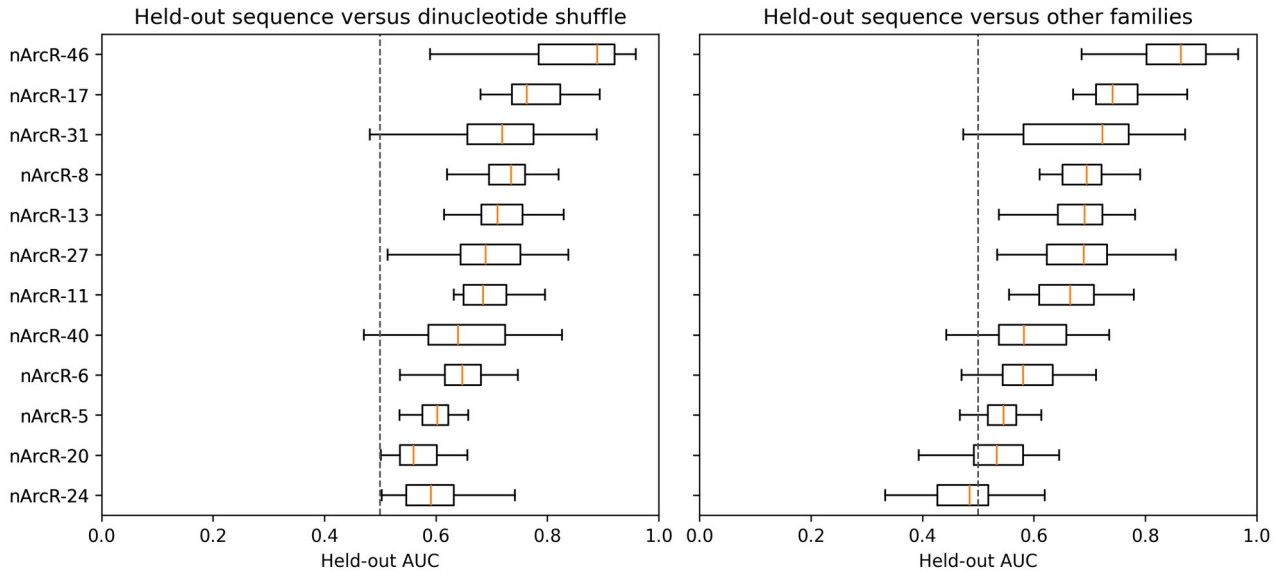
Supplementary Figure S1. Within-panel fixed-family retention under three alternative sampling schemes. Boxes summarize 500 seeded replicates. This analysis asks whether already-defined families remain represented; it is not a de novo clustering sensitivity analysis.

Supplementary Figure S2



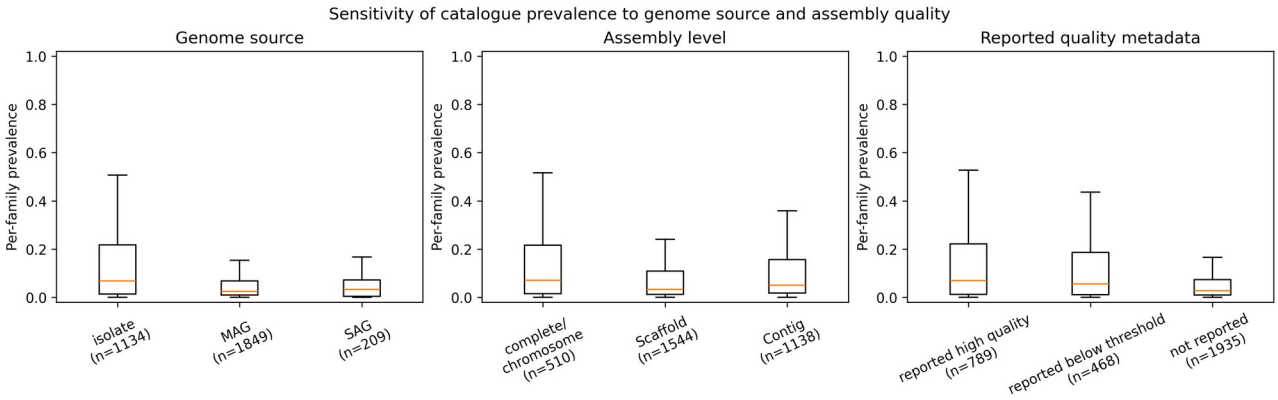
Supplementary Figure S2. Projection of released family HMMs to 500 lineage-matched samples of 331 independent genome accessions. The discovery/projection accession overlap is zero. Results test HMM transfer, not independent family discovery.

Supplementary Figure S3



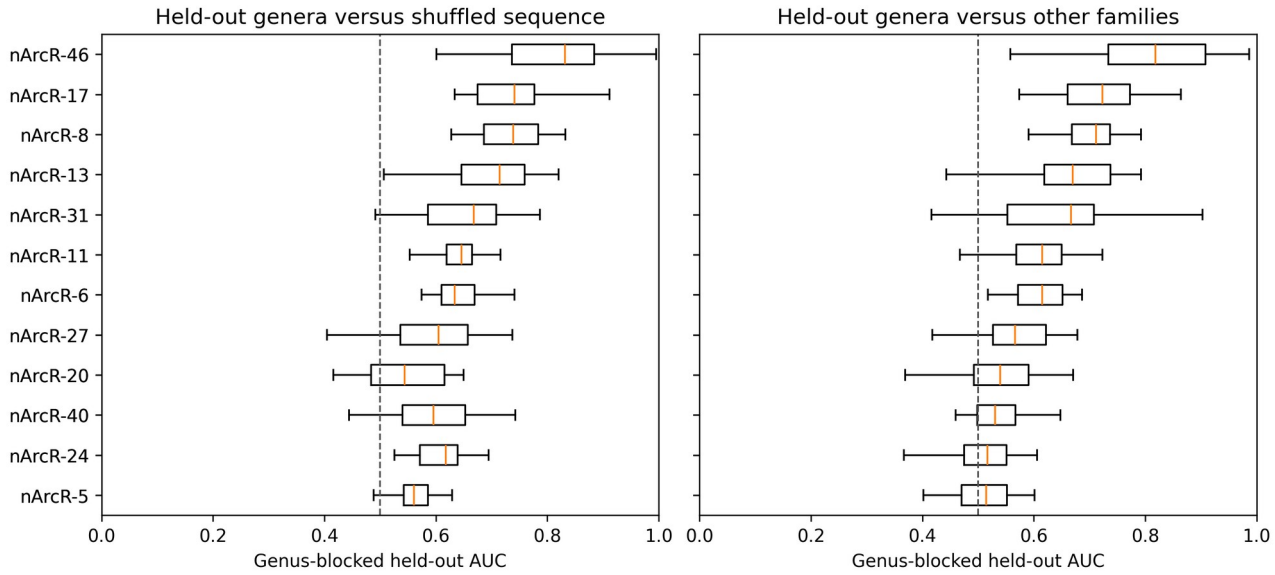
Supplementary Figure S3. Genome-level motif holdout for 12 families. PWMs were fitted only to training genomes and scored against held-out family upstreams, dinucleotide shuffles, and GC-matched other-family upstreams. Repeated splits overlap and are interpreted descriptively.

Supplementary Figure S4



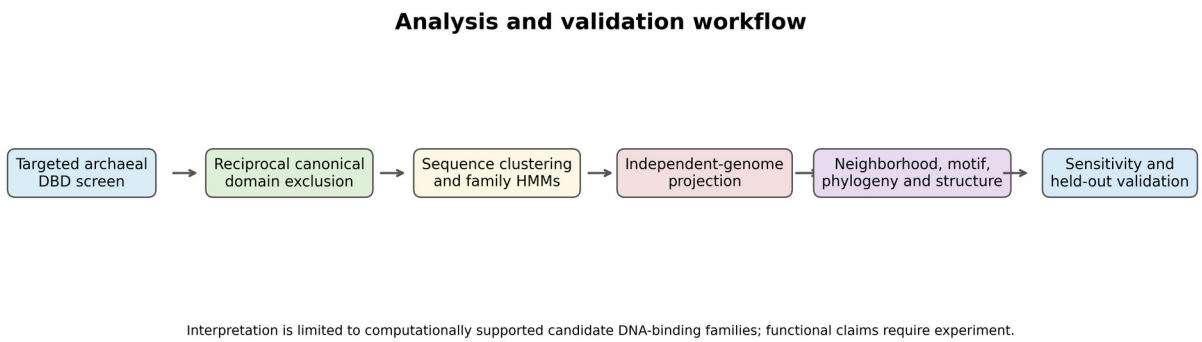
Supplementary Figure S4. Sensitivity of per-family prevalence to genome source, assembly level, and reported completeness/contamination. Boxes summarize the 250 family prevalence values within each stratum. Differences are descriptive because strata are confounded with lineage and sampling.

Supplementary Figure S5



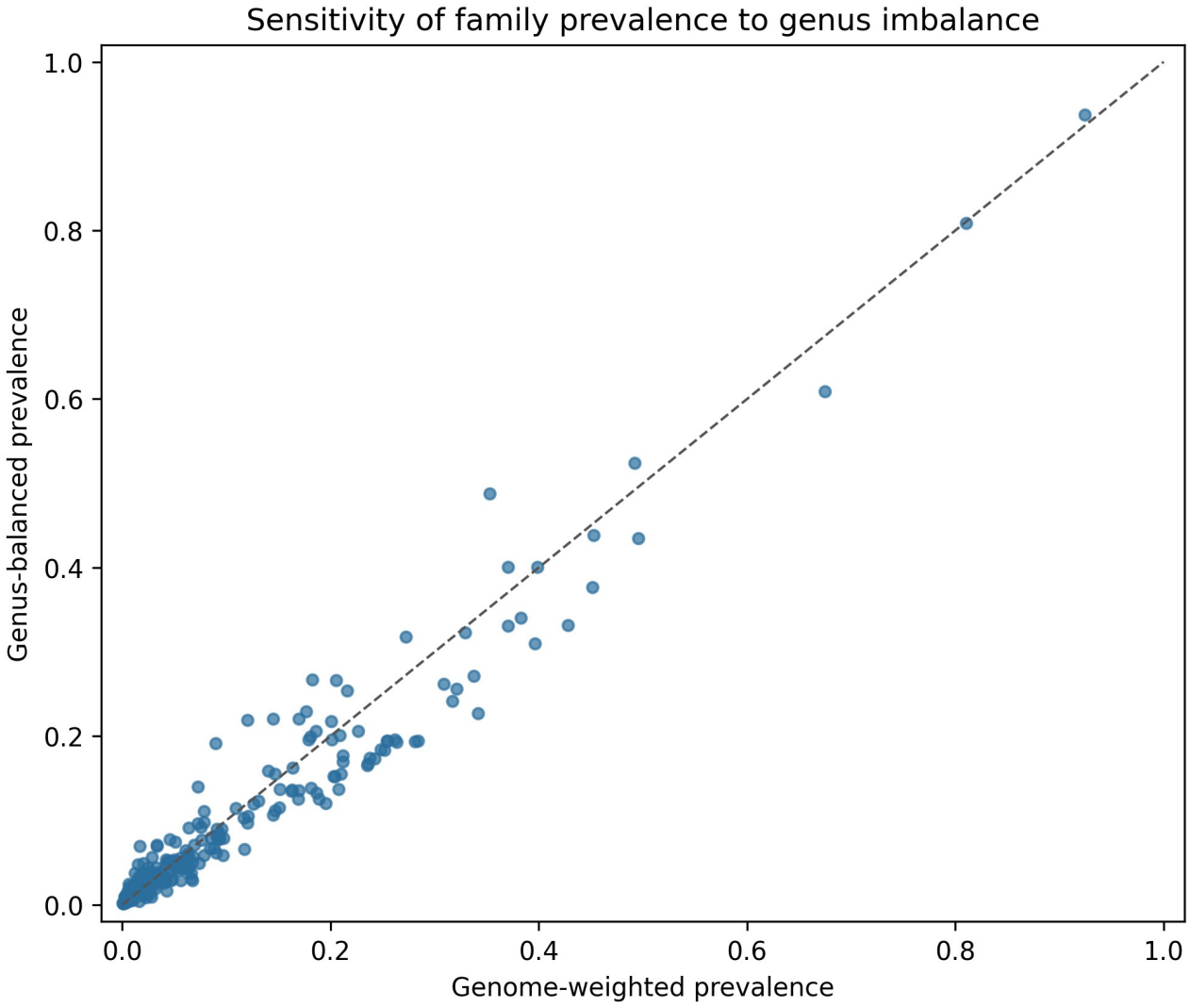
Supplementary Figure S5. Genus-blocked motif holdout with zero train/test genus overlap. Seven of 12 families exceeded AUC 0.5 against other-family controls in at least 26 of 30 resamples. This stricter result shows that some genome-level signal is lineage-associated.

Supplementary Figure S6



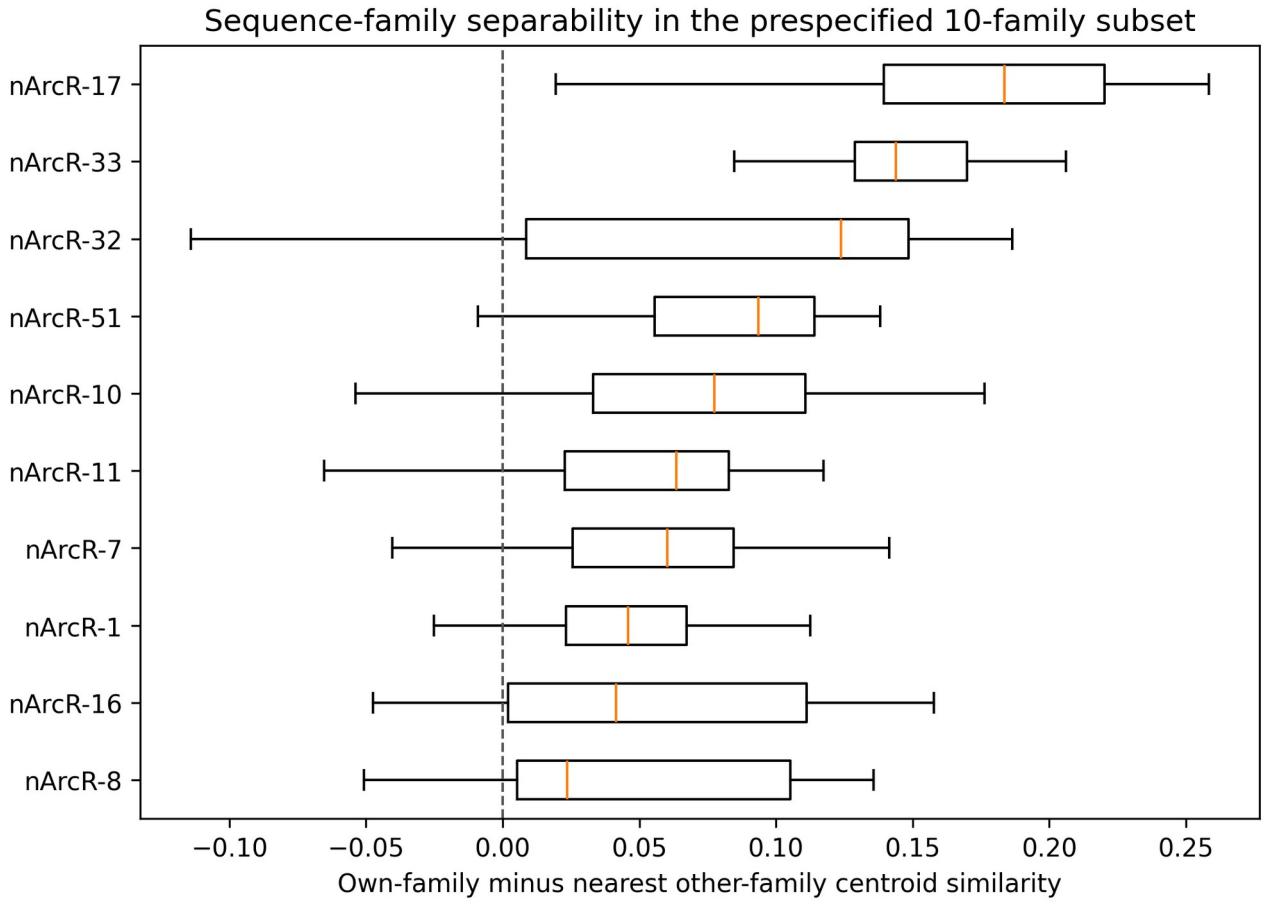
Supplementary Figure S6. Workflow from the targeted two-clan scan through family construction, projection, contextual analyses, and sensitivity validation. The endpoint is a bounded candidate-family catalog, not an experimentally confirmed regulome.

Supplementary Figure S7



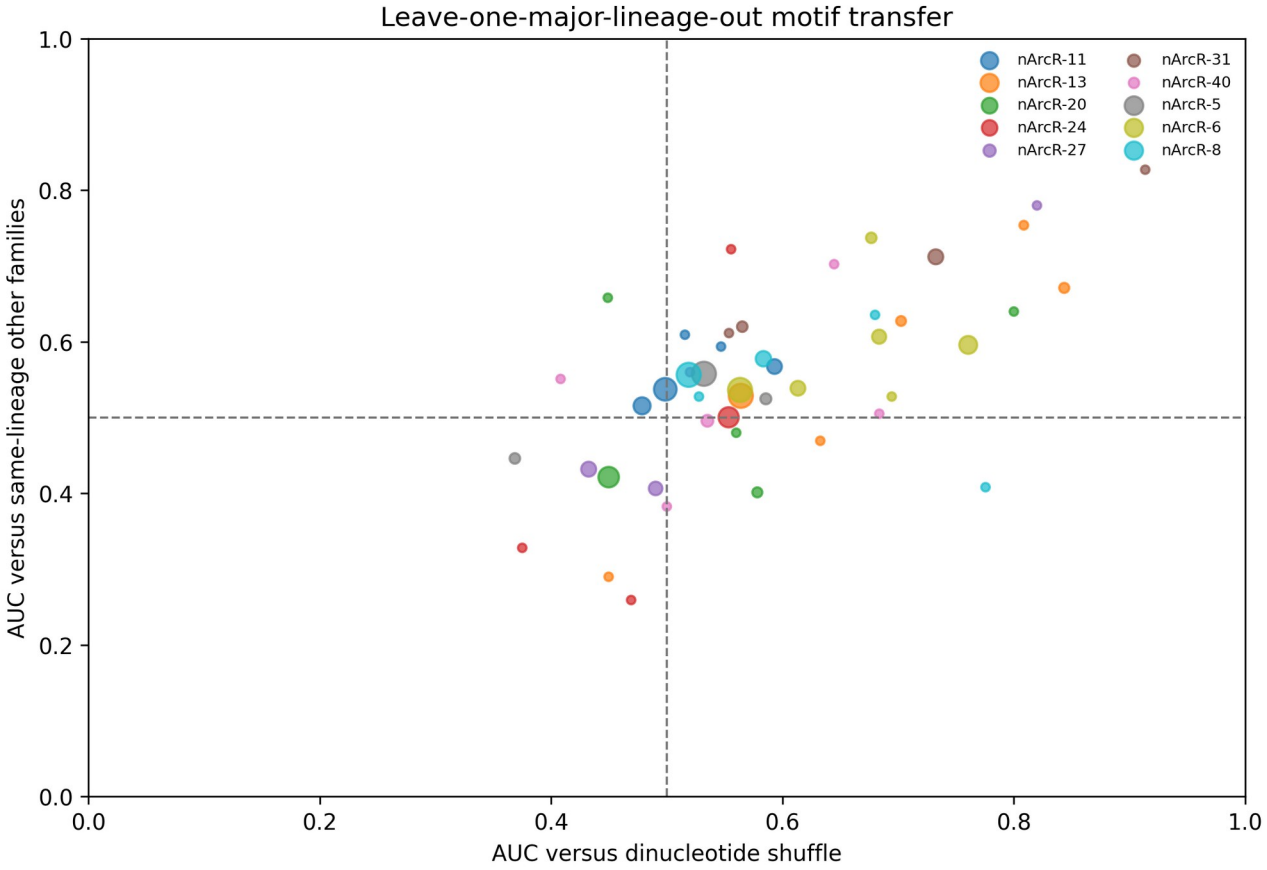
Supplementary Figure S7. Comparison of genome-weighted and genus-balanced prevalence for 250 families. Genus clusters were inferred from deposited organism names. Deviations from the diagonal show sensitivity to uneven genus representation.

Supplementary Figure S8



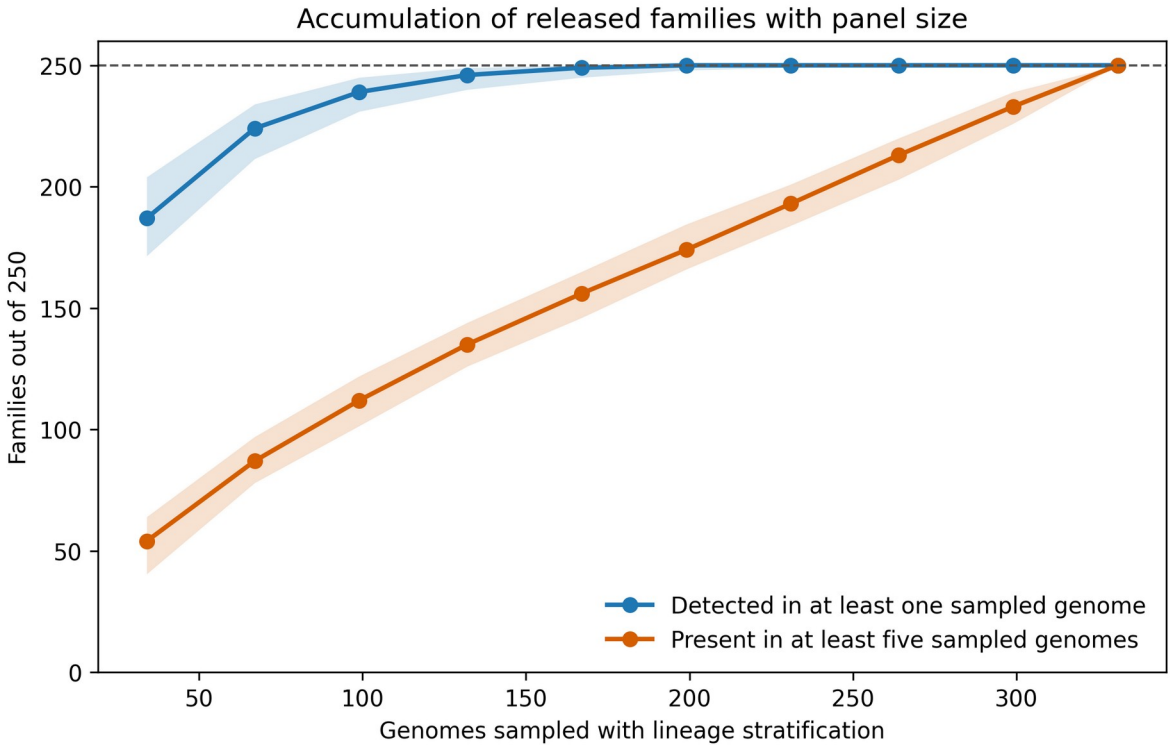
Supplementary Figure S8. Leave-one-sequence-out dipeptide-composition separability for the prespecified ten-family subset. Positive margins favor the sequence's own family over the nearest other-family centroid. Results are not catalog-wide.

Supplementary Figure S9



Supplementary Figure S9. Leave-one-major-lineage-out motif transfer. Point size reflects held-out sequence count. Only blocks satisfying prespecified target, training, and same-lineage control minima are plotted; two families have no evaluable block.

Supplementary Figure S10



Supplementary Figure S10. Lineage-stratified accumulation of the released families as panel size increases. Lines show medians and shaded regions show 95% resampling intervals. The blue curve counts families observed at least once; the orange curve counts families represented in at least five sampled genomes. This is a fixed-catalog sensitivity analysis and does not repeat candidate scanning or de novo clustering.